Speaker script — Track B deck (`trackb.pdf`, 32 pages)

CORRECTION (2026-08-14): the per-beat times below sum to 12:40, not the 9:20 originally claimed — an arithmetic error caught in review. Treat this file as the EXTENDED narration (study/rehearsal version); the timed five-member performance script is `script.md` (blocks M1–M5, 10:20 with cuts). Slides are referenced by TITLE, not page number — pages have shifted three times already tonight and titles have not. Divider pages ("The question", "The instrument", …): say nothing, advance within ~3 seconds; they exist as breathing room, not content. The appendix (A1–A7) is never presented — it is jumped to in Q&A only.

Numbers rule (house law): every number spoken below is in RESULTS.md under the row cited. Do not improvise numbers. If asked something not on a slide: "that's in our results ledger — row so-and-so, I can show you after."

Pronunciation hints - *ancilla* — an-SIL-la - *Hadamard* — HA-da-mar - *AQC* — say the letters: ay-cue-see (Approximate Quantum Compiling) - *Trotter* — TROT-ter - *χ (chi)* — say "kai" - *Loschmidt* — LOSH-mit - *ansatz* — AHN-zahts - *Fermi–Hubbard* — FER-mee HUB-bard

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Title (0:15)

> We are Team 8, the Garbage Collectors. Our topic is the shadow-enhanced Hadamard test — and our story is how we made that test scale, found the trap that would have silently broken it, and then tested our own claim on real hardware. Including the part where the hardware said no.

Outline (0:15)

> Five sections. One thing to know before we start: every tag like R-forty-six on these slides is a row number in our public results ledger — one row per number, with the command that produced it.

The result, up front (1:00)

> Four numbers. Thirty-six times fewer gates for the controlled evolution — as a compilation result. One single-qubit gate to fix a trap that otherwise makes the answer wrong by more than the signal itself. Two QPUs where we tested the gate-count win — and it did not survive; we predicted the numbers before submitting, and we were wrong by twelve times. And one identity that is exactly zero, which gives us an error bar on hardware for free. That is the honest shape of this project: real wins, and real negatives, reported with equal weight. [R046, R054, R044]

Where this sits: the five tracks (0:30)

> The challenge offered five tracks. We completed all five — Track A is the mandatory spectral analyser, twelve out of twelve seeds correct. This deck is Track B, the one we took furthest: past the scaffold, against real baselines, onto two QPUs, and into compilation. [R019, R034]

You compiled U into something cheaper. Is it still right? (0:45)

> Here is the question everything hangs on. Every practical simulation replaces the true evolution U with something cheaper — a Trotter formula, a compiled circuit. The question that decides whether your result means anything is: how wrong is the cheap version, and wrong in *what*? Classically you answer by simulating both — exactly what you cannot do at scale. Track B answers it on the device: put both evolutions on one ancilla and let them interfere.

Anti-control: how the Hadamard test is extended (0:50)

One structural change to the textbook circuit. The sandwich of two X gates around the controlled-W makes W fire when the ancilla is zero, U when it is one. So the ancilla no longer compares U against doing nothing — it compares two dynamics against each other. Everything downstream is unchanged. The algebra is in appendix A-one and A-two if anyone wants it.

Track B: anti-controlled Hadamard test (0:35)

Because this is a new circuit, it gets its own validation, not inherited trust: the identity holds to ten to the minus fourteen on two models. And this check caught a real bug — an endianness mistake in our own reference that passes at t equals zero and fails everywhere else. It never reached the QPU. [R034, R042]

First, what a classical shadow is (0:45)

For anyone new to shadows: measure each qubit in a random basis each shot, keep the record. The randomness is the whole trick — one record set becomes an unbiased estimator for every Pauli observable at once, at a price of three to the weight in variance. The worked table shows the same five shots feeding two different observables.

What the shadows add: delete the final Hadamard (0:45)

Here is our completion of Track B. Delete one gate — the final Hadamard — and the ancilla-tagged shadows split into a sum and a difference of the two dynamics. Add and subtract, and you get every observable under W and under U *separately*, from the same shots. The ancilla says how much the compiled circuit is wrong; the garbage register says *which observable* it got wrong. [R042]

We are not the only way to do this (0:35)

Due diligence: the compiling literature already has verification methods. The Loschmidt echo — apply U, un-apply W, measure. The Hilbert–Schmidt test on double the qubits. We implemented both and verified them before comparing. [R043]

The honest scoreboard (0:40)

And the comparison is not flattering: the echo is cheaper than us at every size — four to six times. If all you want is one number, use the echo. Our arm is justified by what it returns that nothing else does: the phase, and the per-observable breakdown. That is the honest trade. [R043]

AQC makes it scale — crossover at n=4 (0:55)

Now the bottleneck. The standard exact construction of the controlled evolution costs four to the n-plus-one gates — the orange line. Approximate Quantum Compiling grows roughly linearly — the blue line. They cross at four qubits, and by seven qubits AQC is thirty-six times cheaper, at infidelity three times ten to the minus four. Measured, not extrapolated — and it survives a 2D lattice, the Fermi–Hubbard model, and real device routing, which actually widens the gap. [R046, R049, R051, R052]

The trap: a phase-blind compiler (1:00)

But shipping that compression naively destroys the experiment, and nothing warns you. The compiler maximises state fidelity — a magnitude. A Hadamard test is an interferometer — it measures phase. So a compression that converges beautifully returns chi wrong by up to three radians: the error is as large as the signal. We nearly missed it because our own certificate was also a magnitude. The fix costs one single-qubit gate on the ancilla — the phase is free at compile time — and the error drops from one point three to below one percent. We saw the same magnitude-hides-phase failure twice more: once in a teammate's hardware run, once in our own. [R046, R053, R054]

We ran it on a QPU. It failed (1:10)

Then we did the thing you are supposed to do: we tested our own claim. Three versions of the same test, one job, two devices, and the survival predictions written down *before* submitting. Every arm on both QPUs came back as noise — our compiled arm twelve times below its own prediction. And look at the orange arm: its "survival" reads one point four nine — the best score on the slide — and it is unphysical. Chi cannot exceed one. That circuit is simply dead, the ancilla relaxed to zero, and a magnitude-only metric crowns it the winner. So: the gate-count win is real; it does not reach today's hardware at this size; and we can tell you exactly why and what error rate would change that — the arithmetic is in appendix A-seven. [R054, R055]

First Track B on a QPU — the cheap baseline beat us (0:40)

Same discipline on Track B itself: four arms, two devices, one job each. The two-gate echo tracked the exact curve within a few percent; our hundred-and-thirty-six-gate arm did not. Twenty-seven times more accurate, the baseline. It is on the slide because it is true. [R044]

The garbage also ships a free error bar (0:35)

One clean hardware win. This difference of charges is exactly zero — by conservation, for any Trotterisation. So its measured value is pure device error, certified with no reference state and no simulation. It flagged precisely the times where everything else degraded. [R044]

What we claim, and what we do not (0:40)

The summary, in two honest columns. We claim the compilation result, the trap and its fix, the completion of Track B, and the hardware tests themselves. We do not claim the gate-count win reaches hardware — we showed it does not — and we do not claim any advantage over classical computation: the benchmark is small on purpose, so every number could be graded against truth.

Future work: density of states (0:20)

One direction, already verified: feed the same circuit the maximally mixed state and it becomes a density-of-states estimator — the W-equals-identity special case of Track B. [R045]

Reproduce it (0:25)

Everything is public: the repo behind this QR, fifty-plus sourced result rows, the bug log including our retractions, and scripts that reproduce every hardware number in a handful of commands. Thank you — questions welcome.

Timing summary

beat	time	cumulative
Title + Outline	0:30	0:30
Result up front	1:00	1:30
Five tracks	0:30	2:00
The question	0:45	2:45
Anti-control + validation	1:25	4:10
Shadows + delete-H	1:30	5:40
Baselines + scoreboard	1:15	6:55
AQC crossover + trap	1:55	8:50*
running long? see cuts		
QPU failure	1:10	—
Track B QPU + free bar	1:15	—
Claims + future + QR	1:25	**9:20**

The table double-counts nothing; 8:50 is a checkpoint — if you reach the AQC trap slide after 8:00, start cutting.

Cut list, in order (keep the talk at 10:00): 1. *Future work / DOS* (−0:20) — say one sentence over the Claims slide instead. 2. *First, what a classical shadow is* (−0:45) — only if the audience is expert; never cut it for a mixed room. 3. Compress *Baselines* + *Scoreboard* into one beat (−0:30). 4. Never cut: the trap, the QPU failure, the claims slide. They are the talk.

Q&A: use the crib in `SPEAKER_SCRIPT.md` (§Q&A crib sheet) — it covers the n=6-vs-n=7 question, the ratio-vs-absolute question, and the priority-claim fallback. Appendix map for live jumps: A1/A2 anti-control algebra · A3 the two readouts · A4 shadow variance · A5 the combined estimator · A6 DOS · A7 the n=6/n=7 arithmetic.